User Guide

Application: SGLC-Calc: Satellite-Ground Laser Link Budget Tool

Version: V1

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1 Introduction

Satellite-Ground Laser Link Budget Tool is a desktop application for **satellite-ground optical link budget analysis**. It performs comprehensive power budget calculations that account for free-space propagation loss, atmospheric attenuation under various weather conditions (rain, snow, fog, clouds), and atmospheric turbulence scintillation effects.

Key Features

Feature	Description
Link direction	Downlink (Sat-to-Ground) and Uplink (Ground-to-Sat)
Loss modeling	FSPL, pointing loss, optical antenna efficiency
Weather models	Clear Sky / Fog / Rain / Snow / Clouds
Turbulence model	H-V 5/7 C_n^2 profile with ITU-R P.1622-1 standard
Visualization	Three interactive chart views
Export	One-click result export

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2 Getting Started

1. Ensure the required **MATLAB Runtime** is installed (see *Installation Guide*).
2. Launch `SGLC_Calc_V1.exe` (standalone) or open `SGLC_Calc_V1.mlapp` in MATLAB.
3. The first launch may take **30 s – 2 min** for Runtime initialization — this is normal.
4. The main interface of the software will be displayed when the launch is complete.

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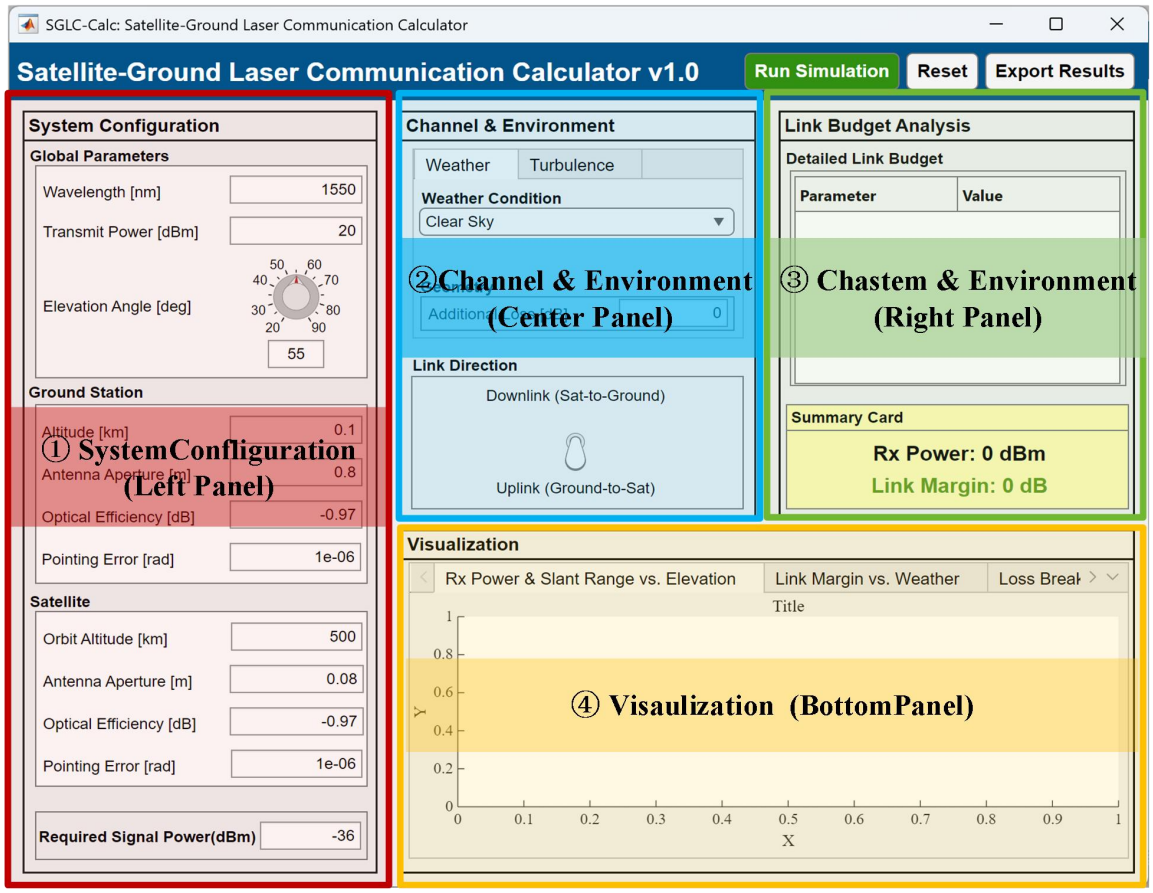
3 Interface Overview

The main window is divided into **four functional panels**:



Toolbar Buttons

Button	Function
Run Simulation	Execute the link budget calculation
Reset	Restore all parameters to defaults
Export Results	Save results to a file

4 Step-by-Step Operation

Step 1 — Configure System Parameters (① Left Panel)

The **System Configuration** panel contains three parameter groups and the receiver sensitivity.

4.1 Global Parameters

UI Label	Description	Unit	Example
Wavelength [nm]	Laser operating wavelength	nm	1550
Transmit Power [dBm]	Transmitter optical power	dBm	20
Elevation Angle [deg]	Link elevation (rotary dial + numeric input)	°	45

> **Tip:** The elevation angle can be adjusted via the **rotary dial** or by typing directly into the numeric field below it. The dial range covers **45°–90°**.

4.2 Ground Station

UI Label	Description	Unit	Example
Altitude [km]	Ground station altitude ASL	km	0.1
Antenna Aperture [m]	Telescope aperture diameter	m	0.8
Optical Efficiency	Optical system throughput (0–1)	dB	−0.97
Pointing Error [rad]	Beam pointing offset	rad	1e-06

4.3 Satellite

UI Label	Description	Unit	Example
Orbit Altitude [km]	Satellite orbital altitude	km	500
Antenna Aperture [m]	Satellite optical aperture	m	0.08
Optical Efficiency	Satellite optical throughput	dB	−0.97
Pointing Error [rad]	Satellite pointing offset	rad	1e-06

4.4 Receiver Sensitivity

At the bottom of the panel, **Required Signal Power (dBm)** defines the minimum detectable power threshold used to compute the link margin (example: `-36`).

Step 2 — Set Channel & Environment (② Center Panel)

4.5 Link Direction

Select the communication direction at the bottom of the center panel:

Option	Meaning
Downlink (Sat-to-Ground)	Satellite transmits → Ground station receives
Uplink (Ground-to-Sat)	Ground station transmits → Satellite receives

4.6 Weather Tab

Switch to the **Weather** tab and select a weather condition from the **Weather Condition** dropdown:

Weather Type	Additional Parameter	Unit
Clear Sky	Direct atmospheric attenuation	dB
Fog	Visibility	km
Rain	Rainfall rate	mm/h
Snow	Snowfall rate	mm/h
Clouds	*(computed automatically)*	—

After selecting a weather type, enter the corresponding parameter value in the input field below.

4.7 Turbulence Tab

Switch to the **Turbulence** tab to configure the atmospheric turbulence model:

1. Toggle **Turbulence Model** to `On` or `Off`.
2. When enabled, enter the **RMS Wind Speed (v) [m/s]** (example: `21`).

3. The label **H-V 5/7** indicates that the Hufnagel-Valley 5/7 C_n^2 profile is used.

Step 3 — Run the Simulation

1. Verify all parameters have been entered.
2. Click the **Run Simulation** button (green) in the toolbar.
3. Results appear instantly in the right panel and the visualization area.

Step 4 — Review Results (③ Right Panel)

4.8 Detailed Link Budget Table

A tabular breakdown of all computed parameters:

Parameter	Description	Example
Slant Range	Satellite-to-ground slant distance	682.94 km
Free Space Loss (FSPL)	Free-space propagation loss	254.87 dB
Total Pointing Loss	Combined Tx + Rx pointing loss	11.53 dB
Optical Antenna Efficiency	Combined Tx + Rx optical efficiency loss	1.94 dB
Weather-related losses	Rain / Snow / Fog / Cloud / Clear Sky attenuation	varies
Turbulence Loss	Scintillation loss (0.00 dB when disabled)	0.00 dB

4.9 Summary Card

Two key metrics are displayed prominently:

- **Rx Power** — Received optical power (dBm)
- **Link Margin** — Margin above receiver sensitivity (dB), shown in **green** when positive

Link margin interpretation:

Link Margin	Status	Meaning
> 3 dB	✅ Viable	Sufficient margin; stable communication expected

0 – 3 dB	Marginal	Link is feasible but has limited headroom
< 0 dB	✕ Infeasible	Received power is below sensitivity; system optimization required

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Step 5 — Explore Visualizations (④ Bottom Panel)

The **Visualization** area offers three switchable chart tabs:

Tab 1: Rx Power vs. Elevation

Displays received power as a function of elevation angle:

Element	Description
Blue solid line	Rx Power at each elevation angle
Red dashed line	Receiver sensitivity threshold (labeled with value, e.g., *Sensitivity (−36.0 dBm)*)

The intersection of the two lines indicates the **minimum usable elevation angle**.

Tab 2: Link Margin vs. Weather

Shows link margin variation across elevation angles for the selected weather condition:

Element	Description
Green solid line	Link margin curve
Red dashed line	Safety margin reference (3 dB)
Black dashed line	Critical threshold (0 dB)

The link operates reliably where the green curve remains above the red dashed line.

Tab 3: Loss Breakdown

A bar chart comparing individual loss contributions, useful for identifying the dominant loss factor in the link budget.

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5 Exporting Results

5.1 Using the Export Button

1. After a successful simulation, click **Export Results** (blue button) in the toolbar.
2. Choose a save location and filename in the dialog.
3. Results are saved in a structured file format.

5.2 Alternative Methods

Method	Instructions
Screenshot	Press `Win + Shift + S` to capture the results or charts
MATLAB workspace	When running inside MATLAB, export the `res` struct via command line

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6 Worked Example

LEO 500 km Downlink — Clear Sky

A complete calculation walkthrough using the default parameters shown in the UI:

Configuration:

Parameter	Value
Wavelength	1550 nm
Transmit Power	20 dBm
Elevation Angle	45°
GS Altitude	0.1 km
GS Antenna Aperture	0.8 m
GS Optical Efficiency	−0.97dB
GS Pointing Error	1e-06 rad
Sat Orbit Altitude	500 km
Sat Antenna Aperture	0.08 m
Sat Optical Efficiency	−0.97dB
Sat Pointing Error	1e-06 rad
Required Signal Power	−36 dBm

Link Direction	Downlink (Sat-to-Ground)
Turbulence Model	Off

Results:

Output	Value
Slant Range	682.94 km
Free Space Loss (FSPL)	254.87 dB
Total Pointing Loss	11.53 dB
Rx Power	**−16.07 dBm**
Link Margin	**19.93 dB** ✅

Conclusion:

A link margin of 19.93 dB is well above the 3 dB safety threshold, confirming a reliable communication link under these conditions.

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7 Important Notes

1. **Units must match the UI labels exactly:**

Wavelength → **nm** · Altitude / Orbit Altitude → **km** · Aperture → **m** · Pointing Error → **rad**

2. **Elevation range:** The rotary dial spans **45°–90°**. Using angles significantly below 45° may produce extreme attenuation values due to the very long atmospheric path.
3. **First-launch delay:** The initial startup is slow because the MATLAB Runtime libraries must be loaded — subsequent launches will be faster.
4. **Parameter validation:** Physically unreasonable inputs (e.g., negative efficiency, extremely high rainfall rates) may produce anomalous results. Ensure all values are within realistic ranges.
5. **Antivirus software:** Some antivirus programs may flag the compiled `.exe`. Add the application directory to your antivirus whitelist if necessary.

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8 Appendix — Parameter Quick Reference

Parameter	UI Label	Unit	Typical Range
Wavelength	Wavelength [nm]	nm	800 – 1550
Transmit Power	Transmit Power [dBm]	dBm	10 – 40
Elevation Angle	Elevation Angle [deg]	°	20 – 90
Receiver Sensitivity	Required Signal Power (dBm)	dBm	−60 – −30
GS Altitude	Altitude [km]	km	0 – 5
Sat Orbit Altitude	Orbit Altitude [km]	km	200 – 36000
Aperture Diameter	Antenna Aperture [m]	m	0.05 – 1.0
Optical Efficiency	Optical Efficiency	dB	−∞ – 0
Pointing Error	Pointing Error [rad]	rad	10^{-7} – 10^{-5}
RMS Wind Speed	RMS Wind Speed (v) [m/s]	m/s	5 – 30
Rainfall Rate	Weather Parameters (Rain)	mm/h	0 – 100
Snowfall Rate	Weather Parameters (Snow)	mm/h	0 – 20
Visibility	Weather Parameters (Fog)	km	0.1 – 50

9 Parameter Error Prompt & Troubleshooting

This chapter fully lists all Parameter Error pop-up prompts during the full operation cycle of SGLC-Calc software, including the exact prompt text consistent with the UI pop-up, trigger conditions, valid value ranges, and standardized correction solutions. All parameter specifications are unified with the software's built-in rules and the previous chapters of this user guide.

9.1 Global Parameters Error Prompts

This section covers parameter errors from the Global Parameters group in the left System Configuration panel.

Error Pop-up Prompt Text	Trigger Condition	Correction Solution
Wavelength input error! Valid range: 800~2000 nm (positive number)	Input wavelength value is less than 800 nm, greater than 2000 nm, or is a non-numeric value, negative number, or 0.	Enter a positive numeric value within 800 nm to 2000 nm. The typical recommended value for satellite-ground laser communication is 1550 nm.

Transmit power input error! Valid range: greater than 0 dBm	Input transmit power value is ≤ 0 dBm, or is a non-numeric value.	Enter a positive numeric value greater than 0 dBm. The typical recommended value is 20 dBm, and the common practical range is 10–40 dBm.
Elevation angle input error! Valid range: 20°~90° (too low causes link interruption)	Input elevation angle value is less than 20°, greater than 90°, or is a non-numeric value, negative number.	Enter a positive numeric value within 20° to 90°. The rotary dial supports quick adjustment from 45° to 90°; values below 45° can be entered directly in the numeric input field. Note that excessively low elevation angles will cause a sharp increase in atmospheric path attenuation and may lead to link interruption.

9.2 Ground Station Parameters Error Prompts

This section covers parameter errors from the Ground Station group in the left System Configuration panel.

Error Pop-up Prompt Text	Trigger Condition	Correction Solution
Ground station altitude input error! Valid range: ≥0 km	Input ground station altitude value is a negative number, or is a non-numeric value.	Enter a non-negative numeric value (≥ 0 km). The typical range for ground station altitude is 0–5 km above sea level.
Ground station altitude input error! Must be less than satellite altitude (current satellite altitude: X km)	Input ground station altitude value is greater than or equal to the currently set satellite orbit altitude.	Enter a value less than the satellite orbit altitude set in the Satellite panel. Simultaneously confirm that the satellite Orbit Altitude parameter is set correctly.
Ground station antenna aperture input error! Valid range: >0 m	Input ground station antenna aperture value is ≤ 0 m, or is a non-numeric value, negative number.	Enter a positive numeric value greater than 0 m. The typical practical range is 0.05–1.0 m, and the recommended default value is 0.8 m.
Ground station efficiency (dB mode) input error! Valid range: ≤ 0 dB (0% - 100%)	Input ground station optical efficiency value is greater than 0 dB, or is a non-numeric value.	Enter a numeric value ≤ 0 dB. Optical efficiency in dB mode corresponds to a system throughput rate of 0% to 100%, where 0 dB equals 100% no-loss efficiency. The typical recommended value is -0.97 dB.
Ground station pointing error input error! Valid range: >0 and ≤1e-3 rad	Input ground station pointing error value is ≤ 0 rad, greater than 1e-3 rad, or is a non-numeric value.	Enter a positive numeric value within (0, 1e-3] rad. The typical practical range is 1e-7–1e-5 rad, and the recommended default value is 1e-6 rad.

9.3 Satellite Parameters Error Prompts

This section covers parameter errors from the Satellite group in the left System Configuration panel.

Error Pop-up Prompt Text	Trigger Condition	Correction Solution
Satellite altitude input error! Minimum LEO satellite altitude: ≥100 km	Input satellite orbit altitude value is less than 100 km, or is a non-numeric value, negative number.	Enter a numeric value ≥ 100 km. The typical value for LEO satellites is 500 km, and the supported range is 200–36000 km for different orbit types.
Satellite antenna aperture input error! Valid range: ≥0.01 m (minimum practical)	Input satellite antenna aperture value is less than 0.01 m, or is a non-numeric value, negative number.	Enter a numeric value ≥ 0.01 m. The typical practical range is 0.05–1.0 m, and the recommended default value for LEO satellites is 0.08 m.
Satellite efficiency (dB mode) input error! Valid range: ≤ 0 dB (corresponds to 0%-100%)	Input satellite optical efficiency value is greater than 0 dB, or is a non-numeric value.	Enter a numeric value ≤ 0 dB. Optical efficiency in dB mode corresponds to a system throughput rate of 0% to 100%, where 0 dB equals 100% no-loss efficiency. The typical recommended value is -0.97 dB.

9.4 General Parameter Error Handling Steps

Follow the standard steps below to resolve any Parameter Error prompts quickly:

1. Click the **OK** button in the pop-up window to close the Parameter Error prompt.
2. Check the parameter name, valid range, and limit conditions in the error prompt, and locate the corresponding parameter input box on the software UI.
3. Refer to the typical value range in **Chapter 8 Appendix — Parameter Quick Reference** of this guide, and enter a valid value that meets the requirements.
4. For associated parameter errors (e.g., conflict between ground station altitude and satellite altitude), correct the values of the associated parameters synchronously.
5. After all parameters are corrected, click the **Run Simulation** button again to execute the link budget calculation.
6. To quickly restore all parameters to the default valid values, click the **Reset** button in the toolbar, then reconfigure the parameters as needed.

10 Software Development Roadmap
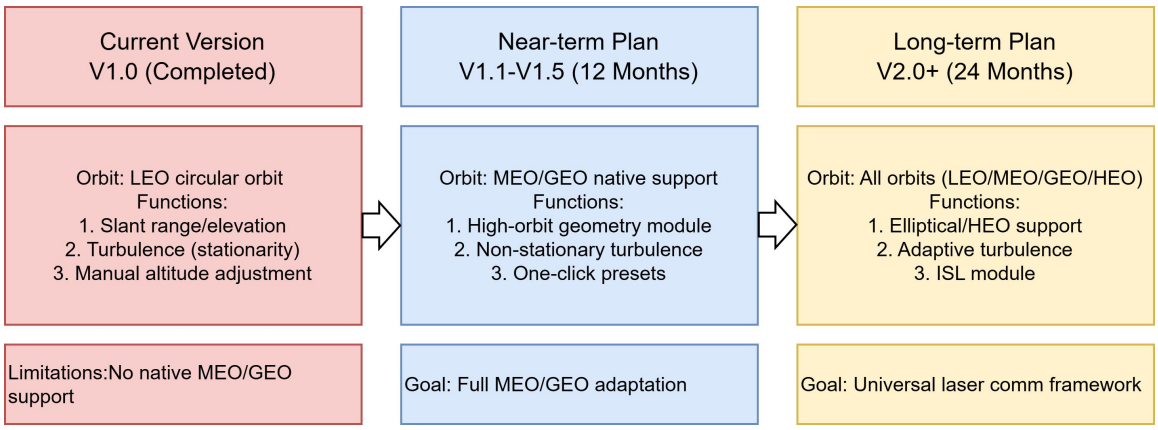
Figure presents the three-phase development roadmap of SGLC-Calc, covering the completed V1.0, near-term V1.1 – V1.5 (12 months), and long-term V2.0+ (24 months). The current V1.0 focuses on LEO circular orbit scenarios, with core functions including slant range and elevation angle calculation, stationary turbulence simulation, and manual altitude adjustment, while lacking native support for MEO/GEO orbits. The near-term plan will prioritize native MEO/GEO compatibility, integrating a high-orbit geometry module, non-stationary turbulence models, and one-click presets to achieve full MEO/GEO adaptation. The long-term plan emphasizes building a universal laser communication framework, extending support to all orbit types (LEO/MEO/GEO/HEO), and introducing adaptive turbulence simulation and an Inter-Satellite Link (ISL) module to cover a broader range of satellite communication scenarios.